

Adversarial ESSD Review of the ITHACA Probabilistic P-E Ensemble

Repository-grounded assessment

This review treats the **current** `master` **implementation** of `imarkonis/ithaca_dataset` as authoritative where the code and internal documentation disagree. That distinction matters. The current workflow is more developed, and in several places materially different, from `docs/methodology_long_v1.1.md` and `docs/figures_plan.md`: the weighting code currently defines **ten** scenarios rather than eight; `04a` generates a 100-member common-random-number sensitivity ensemble plus a 1000-member `change` selection ensemble; and `04c_create_gridded_mc_ensemble.R` is now implemented and generates 100 public gridded members for both `base` and `change`.

The scientific design is nevertheless clear. Five candidate P-E “worlds” are used: ERA5-Land, FLDAS, MERRA-2, TerraClimate, and a deliberately constructed MSWEP-P/GLEAM-E pair. Candidate P and E products are evaluated at grid-cell scale against candidate-specific leave-one-out multi-product reference ensembles. Grid-level performance scores are transformed into weights, screened for physical plausibility, aggregated by IPCC AR6 region × biome, and used as categorical probabilities from which one candidate dataset is drawn for each region × biome and Monte Carlo member. The sampled dataset identity is then used for both P and E and remains fixed over the annual time series of that sampling unit.

The leave-one-out reference is sophisticated enough to remove **literal self-inclusion**, but not genealogical dependence. For each candidate, the code summarizes the remaining reference products by their median climatological mean, median SD, median Sen slope, and majority Mann-Kendall significance behavior. Candidate performance is then measured using relative absolute errors for mean, SD, and slope, together with a binary significance-agreement component.

The current weighting machinery is also more explicit than the manuscript draft implies. The `base` case uses balanced 50:50 blends through the hierarchy. The `change` case gives climatology versus change nominal shares of 25:75 and, within the change component, significance versus slope shares of 75:25. There are also pure/near-endpoint climatology, trend, P-only and E-only scenarios, linear- and exponential-rank alternatives, an exact uniform `neutral` case, and an intentionally inverted adversarial case.

That scenario architecture is a genuine strength for an adversarial assessment: several objections about arbitrary weighting are already testable within the released methodology. But it does **not** solve the more fundamental questions of whether the benchmark constitutes independent evidence, whether the resulting normalized scores have an epistemic probability interpretation, whether the chosen P-E pairs have calibrated uncertainty, or whether the weighted ensemble improves on simply retaining the five originals.

Most importantly, the audit uncovered one implementation issue substantially more serious than the conceptual objections: **the current trend-significance agreement code does not actually require the**

candidate trend to have the same sign as the reference majority. The reference code computes a `majority_sign`, but `03a_dataset_ranking.R` awards significance “agreement” when the reference has a significant coherent trend and the candidate is significant, without comparing the candidate's slope sign with `majority_sign`. A significantly negative candidate can therefore pass against a significantly positive reference majority, and vice versa. This matters particularly because the `change` scenario assigns very large nominal importance to the significance component.

My overall adversarial conclusion is therefore:

The rationale for an ITHACA-like dataset survives scrutiny, but the present implementation cannot yet support a strong claim that ensemble frequencies quantify the relative plausibility of P-E worlds. At present they are best interpreted as **scenario-conditioned sampling weights derived from agreement with a partly dependent multi-product consensus and physical screens**. Independent validation, dependency analysis, and correction of the trend-sign implementation are the main requirements for making a stronger interpretation defensible.

Reviewer attack matrix

Reviewer attack	Papers supporting the attack	Severity	Does it apply to ITHACA?	Best defence
The reference ensemble contains dependent products, so apparent agreement partly counts shared ancestry multiple times. Products can share gauges, forcings, reanalyses, algorithms, or parent products; agreement is therefore not equivalent to independent corroboration.	Vargas Godoy et al. (2025) explicitly identify precipitation-product genealogy and “multiplicity artifacts”, showing that dependence can artificially reduce ensemble variability and distort means. Markonis et al. (2024) likewise relate precipitation disagreement to genealogy. DOLCE explicitly models error covariance because ET products can share forcing/schemes; generalized TCH precipitation fusion similarly treats cross-product error correlation. ¹	Major	Yes, strongly. LOO removes the candidate <i>name</i>, not ancestors or common inputs. The P reference includes gauge products, reanalyses, and derived products with overlapping information histories; the E reference similarly contains products with common meteorological forcings/model families. The five candidate worlds themselves include four same-system P–E pairs plus MSWEP/GLEAM.	Say explicitly that ITHACA estimates agreement-conditioned representativeness, not independent evidential probability. Demonstrate that results are robust after genealogy-aware downweighting or removal of related product families.

Reviewer attack	Papers supporting the attack	Severity	Does it apply to ITHACA?	Best defence
Leave-one-out consensus is still not truth. It rewards conformity. A product can be physically more accurate yet receive low weight if it departs from a biased consensus.	Triple-collocation and generalized TCH literature exists precisely because truth is often unavailable and product agreement alone cannot identify error without assumptions; Xu et al. use estimated error variance–covariance rather than distance to a consensus. Vargas Godoy et al. carefully call their target <i>representativeness</i> and first remove multiplicity artifacts rather than equating the ensemble with truth. ²	Major	Yes. Candidate losses are explicitly distances from median statistics of the other reference products. LOO removes elementary self-circularity but still mathematically rewards membership in the dominant product cluster.	Reframe the benchmark as a consensus/representativeness reference, not observational truth. The defence becomes substantially stronger if independent validation shows that higher consensus-derived weight predicts better held-out performance.

Reviewer attack	Papers supporting the attack	Severity	Does it apply to ITHACA?	Best defence
ITHACA's normalized inverse-loss values are not posterior probabilities that a dataset is correct.	In formal BMA, probabilistic weights have a model-based interpretation and are estimated/calibrated from repeated training data; Raftery et al. describe BMA weights as posterior model probabilities. Knutti et al. emphasize that mapping performance → model quality → weight is itself nontrivial. ³	Major if manuscript uses epistemic probability language; Moderate if wording is corrected	Yes. <code>loss_to_prob()</code> simply computes $1/(\epsilon + L_i)$ and normalizes across candidates. That creates a perfectly valid categorical sampling distribution by construction, but it does not calibrate “probability of being true.”	Use “selection probability,” “sampling weight,” or “scenario-conditioned categorical probability.” State explicitly that frequencies are not Bayesian posterior probabilities, likelihoods, or calibrated probabilities of correctness.
The skill transform is arbitrary and potentially unstable. Inverse relative error strongly rewards near-zero loss; relative trend error can become extreme where the reference slope is close to zero.	Weigel et al. show that incorrect performance weights can perform worse than equal weighting; Knutti et al. and Sanderson et al. emphasize sensitivity of skill weighting to its mapping/strength and to dependence. ⁴	Major	Yes, although substantially anticipated. ITHACA uses $1/(\epsilon + L)$; its own code notes that a near-zero loss can dominate. Relative slope bias divides by <code>abs(ref_slope) + 1e-6</code>, so near-zero reference trends are especially sensitive.	The ten-scenario design is a strong defence: neutral, rank-linear, rank-exponential, inverted, and endpoint weighting schemes deliberately test transform dependence

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Equal weighting may be as good as—or safer than—performance weighting. Why add complexity?	Weigel et al. found equal-weight ensembles can outperform individual members and can be safer than incorrect skill weighting. DOLCE provides the opposite but instructive benchmark: skill weighting improved ET because it used observational training and error covariance. 5	Major	Yes. The current method has a neutral exact 1/n scenario, but the scientific case depends on showing that weighted scenarios add useful information beyond it.	Make neutral/equal weighting the mandatory null benchmark rather than a peripheral sensitivity case. The burden of proof is not necessarily “lower RMSE everywhere”; improved regional discrimination, better calibration, or more informative provenance can also justify the method.

Reviewer attack	Papers supporting the attack	Severity	Does it apply to ITHACA?	Best defence
Why not distribute the five original products and call their spread structural uncertainty?	Multi-product syntheses such as LandFlux-EVAL established that ensemble spread/mean/median can already be scientifically useful; DOLCE shows weighting can improve synthesis, but only when observational constraint and error covariance are explicitly used. Dependence literature also warns that raw spread itself may underestimate uncertainty when members share errors. ⁶	Major	Partly. ITHACA does add something the five raw global products do not: spatial recombination, explicit provenance, locally varying source preference, alternative weighting rules, and a distribution over source mosaics. But those additions require evidence that they improve inference rather than simply repackaging the same five products.	Define the added value narrowly: conditional spatial dataset-choice uncertainty and provenance-preserving recombination, not creation of new observational information.

Reviewer attack	Papers supporting the attack	Severity	Does it apply to ITHACA?	Best defence
For linear quantities, Monte Carlo adds no expected-value information over the deterministic weighted mean.	This follows mathematically from the implementation rather than requiring an external precedent. If $D \sim \text{Categorical}(w)$ and $X = X_D$, then $E[X] = \sum_i w_i X_i$. ITHACA already implements the corresponding deterministic weighted scenario means in 07a.	Moderate	Exactly. For expected P or E, the MC mean converges to the deterministic weighted mean. Therefore Monte Carlo should not be sold as improving the expected estimate.	The real value of MC is elsewhere: discrete provenance, coherent source identity within a sampling unit, nonlinear derived quantities, distribution tails, sign probabilities, and downstream propagation through analyses that are not linear in P/E. 04c additionally stores <code>source_dataset_id</code> making provenance explicit.
Region × biome sampling is an arbitrary spatial support choice that can impose artificial boundaries and suppress within-unit heterogeneity.	Vargas Godoy et al. find that which precipitation dataset is considered representative depends strongly on how the spatial domain is defined, rather than simply its scale. Markonis et al. show sharply environment-dependent product disagreement. 7	Major	Yes. Grid-cell performance is averaged to region×biome weights, after which one dataset identity applies to the entire unit in each member. Local variations in candidate ranking inside the unit no longer affect that member's source choice. Adjacent units can select different sources and introduce source-boundary discontinuities.	This is defensible as a deliberate regularization/coherence scale that avoids salt-and-pepper grid-cell switching. But "physically meaningful" does not mean uniquely correct.

Reviewer attack	Papers supporting the attack	Severity	Does it apply to ITHACA?	Best defence
The “coherent P–E pair” design may preserve correlated biases rather than physical truth. Same-system P and E are not independent pieces of evidence.	DOLCE explicitly treats covariance generated by shared forcings as a source of non-independence. Recent global water-balance evaluation finds low consistency among state-of-the-art P, ET, runoff and soil-moisture datasets and deliberately uses independently derived combinations to diagnose it. ⁸	Major	Yes, but this is a design trade-off rather than an obvious error. ERA5-Land, FLDAS, MERRA-2 and TerraClimate supply same-system P/E; MSWEP P is deliberately paired with GLEAM E. The same sampled identity determines both fluxes.	The good defence is not “coherent pairs are more correct.” It is that they preserve a realizable within-system P–E dependence and avoid arbitrary cross-product combinations. Independent P×E combinations also have a cost: they can manufacture multivariate states no source system ever produced.

Reviewer attack	Papers supporting the attack	Severity	Does it apply to ITHACA?	Best defence
The trend component is unusually fragile, yet receives substantial weight. Global ET products disagree strongly on trend sign and magnitude, and trend significance is sensitive to serial dependence and record length.	Kim, Anabalón & Sharma (2021) found wide differences in sign/magnitude among 11 ET products and rare high concurrence in significant trends. Ruan et al. (2025) found 13 global ET products failed the observed trend sign in >30% of catchments and >40% of flux sites. Serial correlation can materially distort Mann–Kendall significance if untreated. ⁹	Potentially fatal in the current implementation	Yes — and more seriously than the manuscript currently indicates. The code uses Sen slope and Mann–Kendall at $p < 0.1$, and the <code>change</code> hierarchy nominally puts 75% of P/E evaluation into change, of which 75% is significance.	Trend weighting could be defensible after robustness testing, but it cannot presently carry strong evidential meaning.
The current trend-significance “agreement” implementation is incorrect. Opposite-sign significant trends can receive full agreement credit.	Literature establishes why direction matters: cross-product trend sign disagreement is itself a major uncertainty, not a secondary detail. ¹⁰	Potentially fatal if not fixed before submission	Yes, directly in code. <code>majority_sign</code> is calculated upstream, but <code>03a</code> tests candidate significance rather than whether <code>sign(candidate_sen_slope) == majority_sign</code>.	There is no rhetorical defence. This is an implementation defect not a philosophical choice.

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The reference statistics are not estimated over a common temporal support. Dataset climatology, variability and trend may therefore differ partly because the products cover different subsets of 1982–2021.	Long-term trend studies emphasize the substantial sensitivity and disagreement of ET trends even under common evaluation periods. ¹⁰	Major	Yes. Cropping explicitly permits products that do not span the full period. The metadata include, for example, GPCC ending in 2020 and ETMonitor having a much shorter record, while ^{01f} computes product means/SD from whatever finite years are available and requires only ≥ 35 years for trend estimation.	None sufficient without analysis. Product availability differences should not masquerade as product-performance differences.
The Mann-Kendall significance field is used without explicit treatment of serial dependence or spatial multiple testing.	O'Brien et al. document distortion of trend tests by serial correlation. Wilks shows that thousands of simultaneous gridded significance tests can be seriously overinterpreted without field-significance/FDR treatment. ¹¹	Major for significance-weighted products	Yes, with nuance. ITHACA uses local p-values as a weighting feature rather than merely publishing stippled significance maps, so FDR is not a perfect one-to-one requirement. Nevertheless, an uncorrected binary p-value threshold across a massive grid becomes an influential data-processing decision.	Present significance as one heuristic feature, not a verified truth label.

Reviewer attack	Papers supporting the attack	Severity	Does it apply to ITHACA?	Best defence
The P/E physical screen can reject hydrologically legitimate states. Annual ET can exceed annual P under storage depletion, groundwater access, runoff, or irrigation.	Greve et al. derive Budyko behavior permitting $E > P$ when additional stored/imported water is available. Wang & Zhou show groundwater-dependent ET and irrigation can yield $E/P > 1$. A global ET evaluation notes that annual ET may exceed annual P in croplands and accordingly avoided a Budyko constraint there.	Major if called a hard physical validity criterion; Moderate if called a conservative plausibility filter	Yes. ITHACA retains candidates only when $P/E \geq 0.9$, meaning it rejects cases where E exceeds P by more than about 11%, irrespective of groundwater, irrigation, storage or lateral fluxes.	Rename this a plausibility/outlier gate, not a universal conservation law. The relatively permissive 0. threshold already helps

Reviewer attack	Papers supporting the attack	Severity	Does it apply to ITHACA?	Best defence
The AET<PET gate is treated more sharply than the uncertainty of PET warrants.	PET is a model-derived atmospheric demand quantity and methods/forcing assumptions can differ substantially; physical-consistency screens are useful but do not transform PET estimates into exact constraints. The broader hydrological-data literature shows substantial cross-product consistency problems even among current state-of-the-art products. ¹³	Moderate	Yes. A candidate survives only if its mean E falls below PET for a strict majority (>6) of 12 PET realizations.	The use of multiple PET products rather than one is a good defence. Describe the gate as consensus-based plausibility screening.

Reviewer attack	Papers supporting the attack	Severity	Does it apply to ITHACA?	Best defence
ITHACA may claim an “uncertainty ensemble” that is much narrower conceptually than total observational uncertainty.	Mueller-era ET ensemble work and later covariance-aware synthesis show that common errors and within-product error are not captured simply by inter-product spread; Vargas Godoy et al. show dependence can artificially reduce apparent spread. ⁶	Major if “uncertainty” is unqualified	Yes. The MC samples dataset identity , not measurement error, retrieval parameters, forcing uncertainty, reference uncertainty, weighting-parameter uncertainty, product genealogy or spatial-classification uncertainty.	Define the estimand precisely as conditional dataset-choice/structural uncertainty within the selected five P-E worlds.
The weighting is estimated in-sample using the same 1982–2021 behavior whose change is subsequently summarized. A trend-consensus weight will naturally pull the resulting ensemble toward the trend consensus.	The contrast with DOLCE is instructive: DOLCE trained weights against observations and performed out-of-sample tests. Formal probabilistic postprocessing similarly relies on calibration/verification. ¹⁴	Major	Yes. This is not “data leakage” in the predictive-machine-learning sense if ITHACA is purely descriptive, but it does prevent in-sample consensus agreement from being offered as independent validation.	Call the product a conditional synthesis of this historical record, not an independently validated estimate, unless held-out evidence is added.

Reviewer attack	Papers supporting the attack	Severity	Does it apply to ITHACA?	Best defence
The gridded fields can inherit artificial source boundaries. One region×biome unit selects one product, its neighbor may select another.	Spatially heterogeneous performance is well established, and Vargas Godoy et al. specifically demonstrate sensitivity to spatial-domain definition. ⁷	Moderate	Yes. Current 04c expands one region×biome source identity to all grid cells in that unit. The actual P/E fields retain the selected product's within-unit grid variability, so the values are not constant within a unit, but source identity changes discretely at unit boundaries.	This is the price of spatial coherence and is preferable to salt-and-pepper independent grid sampling for many applications.
One hundred released gridded members may be inadequate for stable tail quantiles or rare provenance combinations.	This is principally an MC-convergence issue rather than a hydrological-literature dispute. The repository's own figure plan already identifies convergence as required evidence.	Moderate	Yes. The current public base/change code uses 100 members, although a separate change-selection stream has 1000 members.	Unit-level categorical probabilities are known exactly, so MC should not be used where exact mixture probabilities suffice.
Rounding annual P/E to whole millimeters before diagnostics is unnecessary and could affect small trends or significance near zero.	This is an implementation-level objection rather than one requiring a separate literature claim.	Minor to Moderate	Yes. 01b applies round(..., 0) during ingestion, before mean/SD/trend evaluation.	Annual totals are large enough that effects on climatological means may be negligible, but trend rankings close to zero are a different question.

Reviewer attack	Papers supporting the attack	Severity	Does it apply to ITHACA?	Best defence
Alternative weighting rules were ignored.	Weighting literature strongly recommends sensitivity to weighting strength/form. 15	Already addressed methodologically	No, not as stated. Current code contains ten deliberately diverse scenarios, including uniform, rank-based and adversarial weighting. It also uses common random numbers across scenarios so differences are not confounded by independent MC noise.	Make this an explicit strength.
Monte Carlo destroys product provenance.	—	Already addressed	No. Public gridded member files retain <code>source_dataset_id[cell]</code> ; MC selection tables also preserve source identity and the underlying uniform draw.	Highlight provenance as one of the principal differences between ITHACA and a conventional blended field.

Two additional repository issues do not by themselves invalidate the science but should be corrected before review. First, the documentation is currently behind the code: `methodology_long_v1.1` and the figure plan describe an eight-scenario/older ensemble state while `03e` now implements ten scenarios and `04c` is implemented. Second, the manuscript's broad language about “probabilities” and “plausible alternatives” is ahead of what the current validation supports. These are fixable, but an ESSD reviewer is likely to notice such discrepancies because reproducibility and transparent data provenance are central to a data paper.

Top reviewer attacks

The following are the five criticisms I would expect to cause the most difficulty in a technically strong ESSD review, ordered by present reviewer risk rather than by abstract scientific importance.

Top attack — The change-performance metric is currently misimplemented.

A strong reviewer could write:

“The most serious issue I found is not conceptual but computational. The manuscript states that candidate trends are assessed against the reference majority in sign/significance, yet the supplied implementation does not compare the candidate slope sign with the reference `majority_sign`. A candidate with a statistically significant negative trend can therefore be marked as agreeing with a reference ensemble whose significant majority trend is positive. This is especially consequential because trend significance receives substantial weight and is heavily emphasized in the change-oriented weighting scenario. Given the documented instability and sign disagreement among global ET trends, I do not consider any change-weighted ensemble result interpretable until this logic is corrected and all downstream products are regenerated.”

That criticism is correct. The upstream reference calculation explicitly creates `majority_sign`, but the downstream test does not use candidate sign when awarding significance agreement. The literature makes the error scientifically material rather than cosmetic: Kim et al. found strong disagreement in direction among global ET trends, while Ruan et al.'s independent catchment/flux-tower assessment found incorrect trend signs in a substantial fraction of evaluations. ¹⁰

This is **potentially fatal only if left uncorrected**. It is straightforward to fix. A good correction should explicitly require, when a reference majority significant trend exists,

$$\text{sign}(\beta_{\text{candidate}}) = \text{majority_sign}_{\text{reference}},$$

as well as the intended significance condition. A regression test should then construct positive-reference/negative-candidate and negative-reference/positive-candidate toy examples and require both to fail.

The correction should trigger **full downstream regeneration**, because the affected primitive weight enters hierarchical P/E weights, region×biome weights, source selections and public NetCDF members. Merely correcting manuscript prose would not be sufficient.

Second attack — The ensemble consensus contains pseudoreplication, and leave-one-out does not solve it.

A strong reviewer could write:

“The central evidence model assumes that closeness to the remaining products constitutes evidence for a candidate. However, these products are not independent experimental measurements. They share gauges, meteorological forcing, reanalyses, retrieval assumptions, land-surface models and parent products. Removing the candidate itself from a reference ensemble does not remove its relatives. The resulting consensus can therefore count a shared information lineage several times. This concern is particularly difficult to dismiss because closely related recent work by members of the present author group explicitly identifies such dependence as ‘multiplicity artifacts’ and constructs a genealogy-aware precipitation benchmark to avoid them. ITHACA appears to reintroduce precisely the problem that work warns against.”

This is probably the **weakest conceptual link** in the current methodology.

Vargas Godoy et al. show directly that dependence among precipitation datasets can artificially reduce variability and skew an ensemble toward repeatedly represented sources; they explicitly distinguish representativeness from truth and build a benchmark intended to eliminate multiplicity artifacts. ¹⁶ Markonis et al. likewise find that precipitation disagreement and dataset relationships vary systematically across environments. ¹⁷ DOLCE shows what a stronger ET fusion treatment looks like: its optimization uses the products' error covariance precisely because two products sharing forcing or schemes are not independent evidence. ¹⁸

ITHACA's leave-one-out operation is useful but only solves the simplest circularity:

$$\text{candidate} \notin \text{its own reference.}$$

It does **not** establish

candidate information $\perp$ reference information.

The fair defence is therefore not to insist on independence. It is to characterize genealogy explicitly and show how much the outputs change when families are balanced or dependence is penalized.

A particularly persuasive analysis would build a transparent **dependency graph or matrix**: shared gauges, common forcing/reanalysis, common satellite inputs, common land-surface models, direct derivation links, and P-to-E forcing links. Recompute the consensus and ITHACA weights after, for example, allowing each genealogy family one effective vote. If the main probabilities and regional conclusions survive, this criticism becomes much less damaging.

Third attack — “Sampling probability” is being asked to carry more epistemic meaning than the weighting calculation supports.

A strong reviewer could write:

“The manuscript repeatedly invokes probabilities and plausible worlds, but the reported probabilities are normalized inverse discrepancies from an internally defined consensus. I see no likelihood model, Bayesian posterior, proper scoring-rule calibration, repeated verification framework, or empirical reliability analysis establishing that a weight of 0.6 means that a source is twice as plausible as a source with weight 0.3. Normalization makes the numbers sum to one; it does not make them calibrated probabilities of correctness. The paper therefore risks converting a heuristic ranking into apparently objective probabilistic uncertainty.”

This attack is also correct **if** ITHACA uses probability language without qualification.

The contrast with formal Bayesian model averaging is useful. In BMA, the weights arise inside an explicit predictive statistical model and have a defined posterior interpretation, estimated from training observations. ¹⁹ Knutti et al. make essentially the same conceptual point from climate-ensemble weighting: defining a performance distance is easy compared with justifying the transformation from performance to model quality and then to weights. ²⁰

ITHACA's current operation is:

$$s_i = \frac{1}{\epsilon + L_i}, \quad w_i = \frac{s_i}{\sum_j s_j},$$

followed by hierarchical combinations and categorical sampling.

There is nothing statistically invalid about drawing

$$D \sim \text{Categorical}(w_1, \dots, w_5).$$

The issue is **what** w_i **means**. At present its safest definition is:

the scenario-conditioned probability with which the workflow selects candidate i , given the specified reference ensemble, evaluation metrics, physical gates, spatial aggregation and weighting transform.

It is **not presently validated as**:

the probability that candidate i is the true P-E state.

Changing the terminology alone would remove much of this reviewer risk. The manuscript should consistently prefer **sampling weight**, **selection probability**, or **scenario-conditioned categorical probability** and reserve “probability of correctness,” “posterior probability,” and “objective likelihood” for concepts ITHACA does not estimate.

Fourth attack — There is not yet an independent demonstration that all this machinery is better than the five source products or neutral sampling.

A strong reviewer could write:

“ITHACA derives its weights from product agreement and then demonstrates an ensemble whose behavior follows those weights. That is an internal-consistency demonstration, not an independent validation. I do not yet see evidence that the resulting ensemble is more useful, better calibrated, more hydrologically coherent, or more accurate than (i) the best individual source, (ii) the simple five-product mean, (iii) the raw five-product range, or (iv) the exact equal-probability ensemble. Given that the method adds substantial complexity, ESSD publication requires a clear empirical demonstration of what users gain.”

This is perhaps the attack **most likely to be made by an ESSD reviewer**, because ESSD's interest is not simply whether a clever synthesis can be produced, but whether the resulting data record is demonstrably useful and well characterized.

The literature makes the outcome genuinely uncertain rather than predetermined. Weigel et al. show why equal weighting is a serious null model: inappropriate skill weighting can lose more information than optimal weighting gains, especially for noisy quantities such as precipitation. ²¹ Conversely, DOLCE demonstrates that properly trained, covariance-aware ET weighting can outperform its component products—but that improvement was supported by flux-tower training and out-of-sample testing. ¹⁸

ITHACA has already done the important engineering work needed to answer this criticism: **neutral** is an exact $1/n$ benchmark, multiple transformations are available, and deterministic weighted means can be generated analytically. What remains is empirical validation.

The manuscript should therefore not ask, “Does weighting change the result?” It should ask the harder question:

Does weighting make the result demonstrably more useful?

That means comparing candidate worlds, unweighted mean, neutral sampling, base sampling, change sampling, and deterministic weighted means against the **same held-out benchmarks**.

Fifth attack — Region × biome weighting may create the spatial structure that the paper then interprets as meaningful uncertainty.

A strong reviewer could write:

“The choice of IPCC-region × biome as the atomic sampling unit is consequential but insufficiently validated. Grid-cell skill is first averaged over this unit and then a single product identity is imposed across the entire unit. This suppresses local differences in performance and introduces categorical source boundaries exactly at the chosen classification boundaries. Recent precipitation work shows that dataset representativeness is highly sensitive to how spatial domains are defined. I therefore cannot tell whether ITHACA’s spatial coherence is hydrologically emergent or partly an artifact of the chosen partition.”

This is a serious criticism because the closest relevant study provides unusually direct evidence. Vargas Godoy et al. report that the representative precipitation product is strongly affected by **the definition of spatial domains**, rather than simply domain size. ¹⁶ ITHACA then intentionally converts grid-cell evidence into region×biome probabilities and imposes one dataset choice across each such unit.

The design has a rational purpose: independent grid-cell draws would generate spatially incoherent salt-and-pepper source fields. A region×biome atom acts as spatial regularization.

But regularization entails an assumption:

within-unit gain in coherence > within-unit loss of local adaptivity.

That assumption is not currently demonstrated.

One targeted sensitivity analysis would go a long way: recompute a manageable subset under, for example, IPCC-region-only, region×biome, hydroclimatic zone, and locally smoothed grid-weight supports. Compare dominant source, effective number of datasets, P–E spread, boundary discontinuities and validation skill. If the large-scale uncertainty patterns persist, region×biome becomes a defensible pragmatic scale rather than an arbitrary one.

What the ITHACA ensemble does and does not represent

The ensemble interpretation should be tightened before submission because this is where otherwise reasonable methodology could be overstated.

The current Monte Carlo experiment represents a random variable of approximately this form:

$$D_{r,b} \sim \text{Categorical}(w_{1,r,b}, \dots, w_{5,r,b} \mid \mathcal{R}, \mathcal{M}, \mathcal{G}, \mathcal{S}),$$

where $D_{r,b}$ is the chosen source for region r , biome b ; $\mathcal{R}$ denotes the chosen reference products, $\mathcal{M}$ the evaluation metrics, $\mathcal{G}$ the physical gates, and $\mathcal{S}$ the weighting scenario. The chosen identity then provides the corresponding candidate P and E fields through time. That source provenance is retained explicitly.

Under that interpretation, ITHACA represents **conditional uncertainty over discrete dataset choice and its spatial recombination**.

It does **not**, without additional modelling, represent all uncertainty in terrestrial P and E.

In particular, it does not explicitly sample:

Uncertainty source	Represented by current MC?	Comment
Choice among the five candidate P-E worlds	Yes	This is the core sampled uncertainty.
Spatial differences in candidate preference	Yes, coarsened	Evidence originates at grid cells but categorical choice occurs at region×biome level.
Weighting-rule uncertainty	Partly	Ten scenarios exist, but public interpretation must keep plausible scenarios distinct from diagnostic/adversarial ones.
Random Monte Carlo sampling error	Yes, but purely numerical	Reducible by more members and not an Earth-system uncertainty.
Measurement error inside each source product	No	A sampled member takes the source value as given.
Retrieval/algorithm parameter uncertainty within products	No , except insofar as it contributes to between-product differences	Product-specific uncertainty distributions are not sampled.
Shared/common-mode error across products	No	Indeed, shared errors can make the five worlds look more concordant than independent evidence would justify. ²²
Uncertainty in the consensus reference itself	Not propagated	Reference medians and majority decisions are treated as fixed when weights are generated.
Product genealogy/dependence uncertainty	No	Exact-product LOO is not genealogy correction.
Candidate-set incompleteness	No	All probabilities are conditional on these five candidate worlds.
Uncertainty in region/biome classification as an appropriate weighting support	No	The classification is fixed.

Uncertainty source	Represented by current MC?	Comment
Uncertainty in physical-screen thresholds	No in the main MC	P/E=0.9 and PET-majority decisions are fixed.
Time-varying relative product skill	No	One source identity is selected per unit/member and retained across years.
Fully general P-E covariance uncertainty	No	P/E covariance is inherited from the five prescribed pairs rather than estimated as a distribution.
Unrepresented hydrological processes such as groundwater, irrigation or lateral redistribution	No explicit component	These processes can additionally affect the plausibility screens. ²³

That distinction should be prominent in the abstract, Methods and data description:

ITHACA ensemble spread is not a total observational uncertainty interval. It is a conditional representation of uncertainty associated with selecting among the included P-E datasets under specified evaluation, screening, spatial-aggregation and weighting assumptions.

That wording is scientifically strong because it is precise. It also prevents users from making a common but serious mistake: interpreting a narrow ITHACA distribution as evidence that terrestrial P or E itself is precisely known. Product dependence can make an ensemble narrow even while its shared error remains large. ²²

There is a second interpretive distinction worth making explicit. **Monte Carlo is not needed to obtain ITHACA's expected P or E field.** Because categorical sampling is used,

$$E[P] = \sum_i w_i P_i, \quad E[E] = \sum_i w_i E_i.$$

The existing deterministic ^{07a} scenario-weighted mean is therefore the analytic expectation of the corresponding categorical ensemble.

This is not an argument against Monte Carlo. It tells the authors what its scientific justification must be.

The MC ensemble is useful where users need:

source-preserving realizations, spatially coherent combinations of product identities, uncertainty propagation through nonlinear operations, probabilities of sign or threshold exceedance, multivariate derived quantities, tail behavior, or downstream models requiring complete alternative P/E realizations rather than a fractional blend. The public files' `source_dataset_id` is particularly valuable here.

Conversely, if the paper ultimately reports almost exclusively ensemble means, then a sceptical reviewer can reasonably conclude that most of the Monte Carlo machinery is unnecessary.

Highest-value tests before submission

Several analyses are necessary, but they are not equally valuable. The ordering below is adversarial: it is based on which test would neutralize the greatest number of possible ESSD objections.

First, fix the trend-sign implementation before doing any scientific sensitivity analysis. The present behavior is demonstrably inconsistent with the intended “sign/significance agreement” concept. No amount of validation of outputs generated with that error can substitute for correcting it.

After that correction, the single analysis with the **largest reviewer-risk reduction** would be a **strictly held-out, multiscale validation experiment comparing ITHACA to all obvious null alternatives**.

A convincing validation design would have three linked levels.

For precipitation, use gauges that are genuinely independent of the products being evaluated wherever such independence can be established. This matters because gauge datasets incorporated in merged products cannot simply be recycled as “external” truth; the dependence problem is exactly why modern precipitation evaluation has become genealogy-conscious. ¹⁶

For evapotranspiration, use withheld eddy-covariance observations and basin water-balance ET. DOLCE demonstrated the value of out-of-sample flux-tower evaluation, while the 2025 global ET trend evaluation combined basin water balance with long-term eddy-covariance records precisely because trend skill cannot be inferred reliably from product agreement. ²⁴

For the coupled P–E variable, use independent basin closure:

$$P - E \approx Q + \Delta S,$$

with quality-controlled discharge and storage-change information where temporal and spatial scales make that comparison defensible. The importance of such a check has become stronger, not weaker: the 2026 global comparison of thousands of P–ET–runoff–soil-moisture combinations found **overall low water-balance consistency** among state-of-the-art hydrological products. ¹³

Each benchmark should compare the exact same alternatives:

- five individual source worlds,
- unweighted five-product mean,
- raw five-product range,
- neutral 1/n ITHACA sampling,
- base weighted mean,
- base categorical ensemble,
- change weighted mean/ensemble where relevant.

The important outputs are not merely RMSE. For a probabilistic data product, reviewers should see **bias, temporal correlation/variability, trend skill where records allow it, water-balance closure, interval coverage, sharpness, and reliability/calibration**. If a nominal 90% ITHACA interval contains an independent reference only 50% of the time, it is underdispersive whatever its internal elegance. If it covers 99% but is no narrower than the raw candidate range, the weighting adds little probabilistic information.

The second-highest priority is a **genealogy/dependence sensitivity experiment**. This should use a transparent table documenting direct and indirect relationships among all P/E reference and candidate products and test at least one dependence-aware alternative. The case is particularly strong because recent precipitation literature identifies exactly these multiplicity artifacts as a source of misleading ensemble inference. ¹⁶ A perfect formal error-covariance estimate is not required to satisfy a reviewer; showing that no major result hinges on repeatedly represented families would already materially strengthen the paper.

The third priority is **trend robustness**. Current global ET literature gives unusually little justification for strong confidence in product trend rankings: multi-product trends disagree in both sign and magnitude, and direct observational evaluations find substantial sign failures. ¹⁰ At minimum, compare:

Sen/MK as presently implemented,
autocorrelation-aware significance,
no significance component,
slope-only change score,
 $p < 0.05$ versus $p < 0.1$,
common temporal-support evaluation.

Because ITHACA evaluates many locations, treatment of significance also needs careful wording; gridded significance fields are particularly vulnerable to overinterpretation under multiple testing. ²⁵

The fourth priority is a **sampling-support sensitivity**. Region×biome should be compared against at least two plausible alternatives. The output should not be a qualitative map alone. Report the fraction of land area where the dominant dataset changes, changes in effective number of datasets, changes in P–E uncertainty, boundary-jump statistics, and validation skill. This is the only convincing answer to the domain-definition result of Vargas Godoy et al. ¹⁶

The fifth is a combined **physical-gate sensitivity**. The paper should quantify how much each gate alters candidate support and final results under:

no gate, P/E gate only, PET gate only, both gates,

plus reasonable threshold perturbations. Excluded cells should be stratified by irrigation, aridity, groundwater dependence where possible, biome and high-latitude environment. This prevents a global “physics filter” from silently excluding exactly the places where terrestrial storage or human water redistribution violates a simple local annual balance. $ET > P$ is demonstrably possible under additional water supply, groundwater access and irrigation. ¹²

Finally, run **MC convergence and analytic expectation checks**. For each scenario, compare MC means against the exact ^{07a} weighted means and demonstrate convergence as member count increases. The repository's own figure plan already correctly identifies this as required evidence. Because unit-level

distributions have only a few categorical states, exact probabilities should be preferred over noisy MC quantiles wherever they can be calculated analytically.

Red-team verdict

What is the weakest conceptual link in ITHACA?

The weakest conceptual link is the move from

“a product agrees with a leave-one-out multi-product consensus”

to

“therefore that product should be sampled more often as a plausible representation of reality.”

Those statements are not equivalent.

The method presently establishes **relative representativeness with respect to a chosen ensemble**. It does not yet establish externally calibrated accuracy or epistemic probability. This distinction becomes particularly important because the reference products are genealogically dependent. Vargas Godoy et al.'s multiplicity-artifact result is almost perfectly targeted at this weakness: repeated/shared information can make consensus appear stronger than the independent evidence warrants. ¹⁶

This does **not** invalidate agreement-based weighting. It dictates the interpretation and burden of validation.

What is the weakest methodological link?

In the **current code**, the weakest methodological link is unequivocally the trend-significance implementation.

The intended metric distinguishes agreement in long-term change, but the code can classify an opposite-sign significant candidate trend as agreement because it checks statistical significance without comparing the candidate's direction against `majority_sign`. Because change receives substantial weight—especially in the `change` scenario—and because global ET trend direction is empirically known to disagree among products, this is not a negligible edge case. ¹⁰

Immediately after that defect comes the lack of common temporal support in reference evaluation.

What claim is most likely to be challenged?

The most challengeable claim is any version of:

“The ensemble probabilities quantify the probability/plausibility that a P-E realization is correct.”

The code does not presently support that calibration. The weights are normalized transformations of agreement scores. Formal probabilistic approaches such as BMA attach probability meaning through an explicit statistical model and calibration data; ITHACA currently does not. ³

A much more defensible claim is:

“ITHACA supplies scenario-conditioned probabilities for sampling among candidate P-E datasets according to their spatially varying agreement with a leave-one-out reference ensemble and physical-plausibility criteria.”

The distinction is not semantic. It determines what users may infer from ensemble frequency.

A second likely challenge is the phrase **“no single input dataset is privileged.”** The method does not privilege one candidate globally, but it absolutely privileges particular **definitions of evidence**: the selected reference products, consensus statistic, metrics, physical gates, region×biome aggregation, inverse-loss transformation and scenario shares. The manuscript should say that no single **candidate source** is prescribed as universally best, rather than implying methodological neutrality. The workflow itself documents substantial structured choices.

What single analysis would remove the most reviewer risk?

After correcting the trend-sign bug, the highest-value analysis is:

A genuinely independent, multiscale held-out validation of the five candidates, neutral ensemble, weighted ITHACA ensemble and deterministic weighted mean, including probabilistic calibration and basin-scale P-E water-balance closure.

That single framework attacks several criticisms simultaneously:

consensus is not truth	→ independent benchmark,
weights are arbitrary	→ test predictive relationship between weight and skill,
why not equal weighting?	→ neutral comparison,
why not raw products?	→ candidate comparison,
P-E pairing may preserve bias	→ water-balance closure,
ensemble is not probabilistic	→ coverage/reliability test,
physical gates may harm performance	→ stratified gate comparison.

It would turn the paper from a sophisticated **internal synthesis** into a validated data product.

The 2026 ESSD water-balance assessment makes such evidence especially timely: modern P, ET, runoff and soil-moisture datasets remain surprisingly inconsistent when confronted with water-balance relationships, and consistency is strongly spatially variable. ¹³

What limitation should be admitted proactively?

The most important limitation statement should be approximately:

ITHACA quantifies uncertainty associated with the choice among a finite set of candidate P–E datasets, conditional on the adopted reference ensemble, evaluation metrics, physical screening, spatial sampling units, and weighting formulation. Ensemble spread should therefore not be interpreted as total uncertainty in terrestrial precipitation, evapotranspiration, or P–E. In particular, common errors and dependencies among products, within-product measurement and retrieval uncertainty, uncertainty in the reference ensemble, and uncertainty arising from candidate-set selection are not fully propagated.

That admission would prevent several reviewer attacks rather than invite them. Dependence-aware ensemble studies already show why inter-product spread cannot automatically be equated with complete uncertainty. ²²

A second limitation should state that region×biome selection is a **spatial regularization choice**: it intentionally trades local adaptability for coherent source identity. That is scientifically defensible, but the literature shows that domain definition itself can affect which precipitation dataset appears representative. ¹⁶

A third should state that the physical gates are **plausibility tests rather than inviolable physical laws**, particularly because $E > P$ can be valid where storage, groundwater, irrigation or lateral water supply contributes to ET. ¹²

Does any criticism genuinely undermine the rationale for publishing ITHACA?

No—not at the level of the core dataset concept.

There is a scientifically legitimate use case for a product that preserves the identity of existing global P–E datasets, converts spatially varying assessments into explicit selection weights, generates spatially coherent alternative source mosaics, and allows downstream analyses to propagate **dataset-choice uncertainty** rather than silently choosing one source. The fact that current hydrological products disagree strongly across environments and have low joint water-balance consistency arguably strengthens the need for transparent uncertainty-aware products rather than eliminating it. ²⁶

What **would** undermine the current submission is a combination of three things:

1. leaving the trend-sign bug in place;
2. calling the normalized consensus weights objective probabilities of truth;
3. providing no independent evidence that the weighted/recombined product adds useful information beyond the originals or neutral sampling.

The first is a correctable implementation defect. The second is solved mainly by conceptual precision and, ideally, calibration. The third requires additional science.

The most dangerous possible validation outcome would be that, after correcting dependencies and trend logic, **neutral 1/n sampling or the unweighted mean performs just as well as ITHACA everywhere, the choice of spatial classification substantially changes source probabilities, and the supposedly**

weighted ensemble is systematically underdispersive against independent observations. Such a result would substantially weaken any claim that performance-derived probabilities are scientifically informative.

Even that result would not necessarily make the dataset unpublishable. It would change what the contribution is. ITHACA could still be valuable as a **provenance-preserving structural-sensitivity ensemble** whose purpose is to expose the consequences of dataset choice rather than to rank worlds by calibrated plausibility.

The strongest defensible scientific position, conditional on fixing the implementation and adding validation, is therefore:

ITHACA should be presented not as an observational posterior or a replacement “best estimate,” but as a provenance-preserving, spatially structured ensemble that propagates conditional uncertainty arising from the choice among existing P-E products, while explicitly exposing sensitivity to the assumptions used to assign those choices.

That formulation retains the genuinely useful part of the method, avoids claiming probabilistic certainty that has not been calibrated, and directly addresses the most serious objections supported by the current literature through **5 October 2026**.

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